

User Guide: Enterprise AI Decision Intelligence Platform

This guide explains how business users and technical users should use the application.

1. What The App Does

The application helps teams review AI/ML decision models, understand their business impact, and inspect the technical evidence behind each run.

It is organized around four model workflows:

- TensorFlow Next Best Action Model
- TensorFlow Churn Propensity Model
- Bayesian Media Mix Optimization
- Causal Incrementality Model

Each workflow moves from model inputs to recommendation, impact forecast, decision evidence, activation output, and governance evidence.

2. Recommended Presentation Flow

Use this sequence for an end-to-end walkthrough:

- Start at AI Decision Portfolio.
- Open the Model Decision Workbench.
- Walk through model inputs, recommendation, impact forecast, and decision evidence.
- Open AI Impact Summary to show portfolio-level health.
- Open AI/ML Platform Architecture to explain the technical runtime and evidence path.

3. Visual Walkthrough

The screenshots below show the recommended flow a user should follow during a business or technical walkthrough.

Flow Step 1: Start With The AI Decision Portfolio

Use this page to introduce the model portfolio and explain the four AI/ML decision workflows.

AI Decision Portfolio

The screenshot displays the 'Enterprise AI Decision Portfolio' dashboard. At the top, there are four tabs: 'AI Decision Portfolio' (selected), 'AI Impact Summary', 'Model Decision Workbench', and 'AI/ML Platform Architecture'. Below the tabs, a dark blue header section contains the title 'Enterprise AI Decision Portfolio' and a subtitle: 'A governed portfolio of AI/ML decision models for customer growth, retention, marketing optimization, and causal measurement, with evidence from model input to activation.' Below this header are three buttons: 'AI Portfolio: Growth + Retention', 'Model Cadence: Weekly / Monthly', and 'Audience: Business + AI Leaders'. A section titled 'How To Present The AI/ML Story' provides instructions: 'Start with the model portfolio, open each AI decision flow, and show how model inputs become governed recommendations, impact forecasts, and activation-ready decisions.' It includes two buttons: 'Open Model Workbench' and 'Open AI Impact Summary'. The main content area features four model cards, each with a title, model evidence, description, model family, decision output, primary metric, and an 'Open Model Workflow' button. The models are: 1. 'TensorFlow Next Best Action Model' (Model evidence: Latest run) - Scores eligible customer actions and channels by expected uplift, confidence, policy limits, and fatigue risk. Model family: TensorFlow uplift decisioning. Decision output: Ranked activation action pack. Primary metric: 30-day incremental retention lift. 2. 'TensorFlow Churn Propensity Model' (Model evidence: Latest run) - Predicts churn risk, prioritizes save treatments, and estimates retention impact for high-risk customers. Model family: TensorFlow risk + save strategy. Decision output: Prioritized retention treatment plan. Primary metric: 60-day incremental churn reduction. 3. 'Bayesian Media Mix Optimization' (Model evidence: Latest run) - Uses Bayesian marketing science to evaluate response curves, seasonality, saturation, and budget movement. Model family: PyMC-style Bayesian MMM. Decision output: Channel reallocation plan. Primary metric: Incremental revenue at fixed budget. 4. 'Causal Incrementality Model' (Model evidence: Latest run) - Separates true campaign lift from baseline demand and recommends scale, pause, or retest decisions. Model family: EconML/DoWhy-style causal inference. Decision output: Campaign scale/pause/retest policy. Primary metric: Incremental conversions per campaign. At the bottom, a section titled 'Recommended AI/ML Presentation Sequence' lists: '1. Start with TensorFlow Next Best Action to show customer-level AI decisioning.'

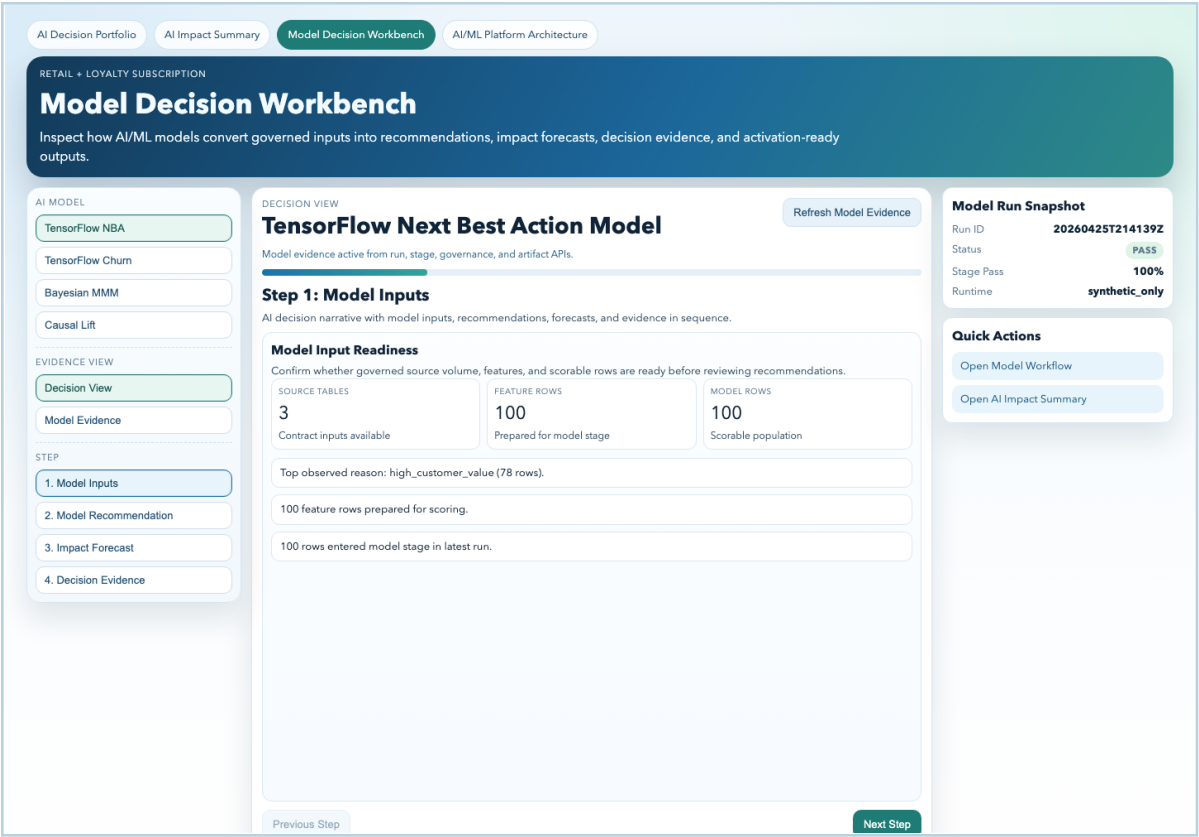
What to point out:

- Four governed AI/ML decision models.
- Each model has a clear business purpose and primary metric.
- The recommended presentation sequence starts with customer-level AI decisioning.

Flow Step 2: Open The Model Decision Workbench

Open a model workflow and start with Model Inputs. This is where users see the data readiness and scorable population before looking at recommendations.

Model Inputs



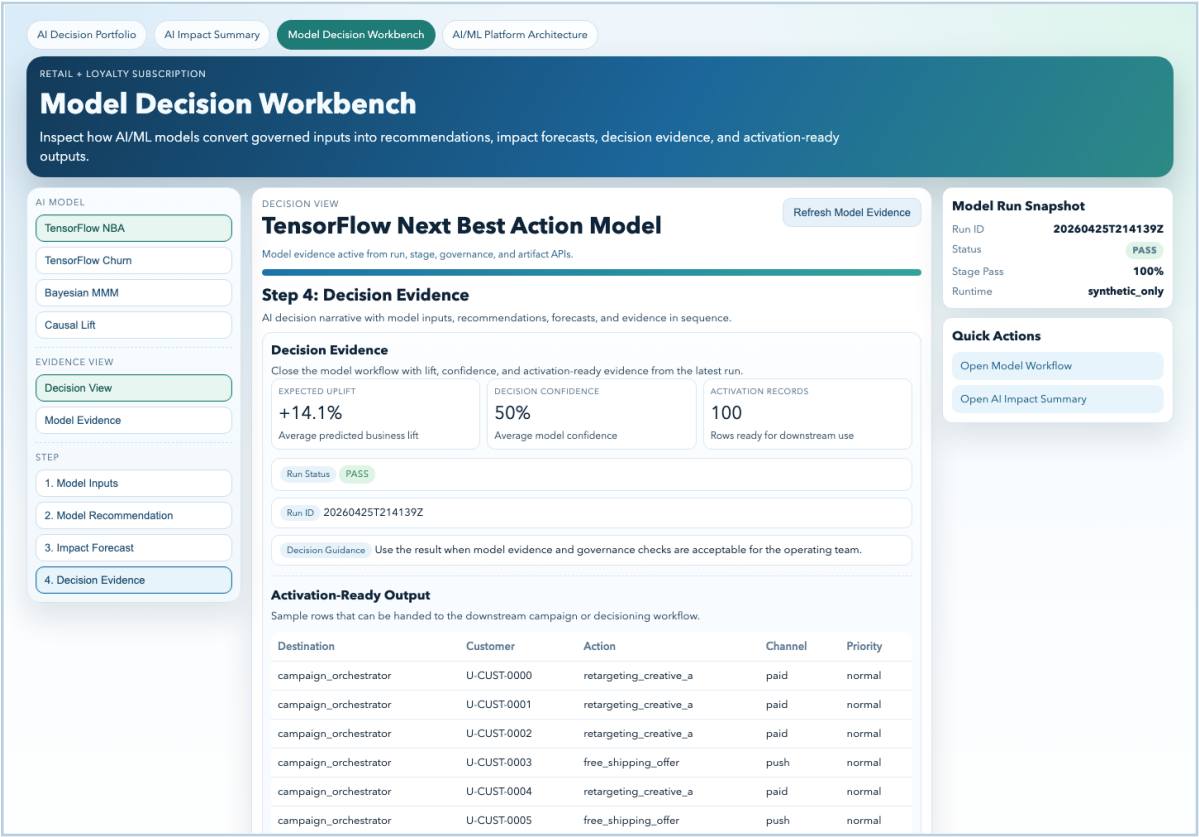
What to point out:

- The left navigation keeps model, evidence view, and step selection visible.
- The center panel explains one decision step at a time.
- The right-side snapshot confirms run ID, status, stage pass rate, and runtime.

Flow Step 3: Review Decision Evidence

Move to Decision Evidence to close the business workflow with uplift, confidence, activation records, and run status.

Decision Evidence



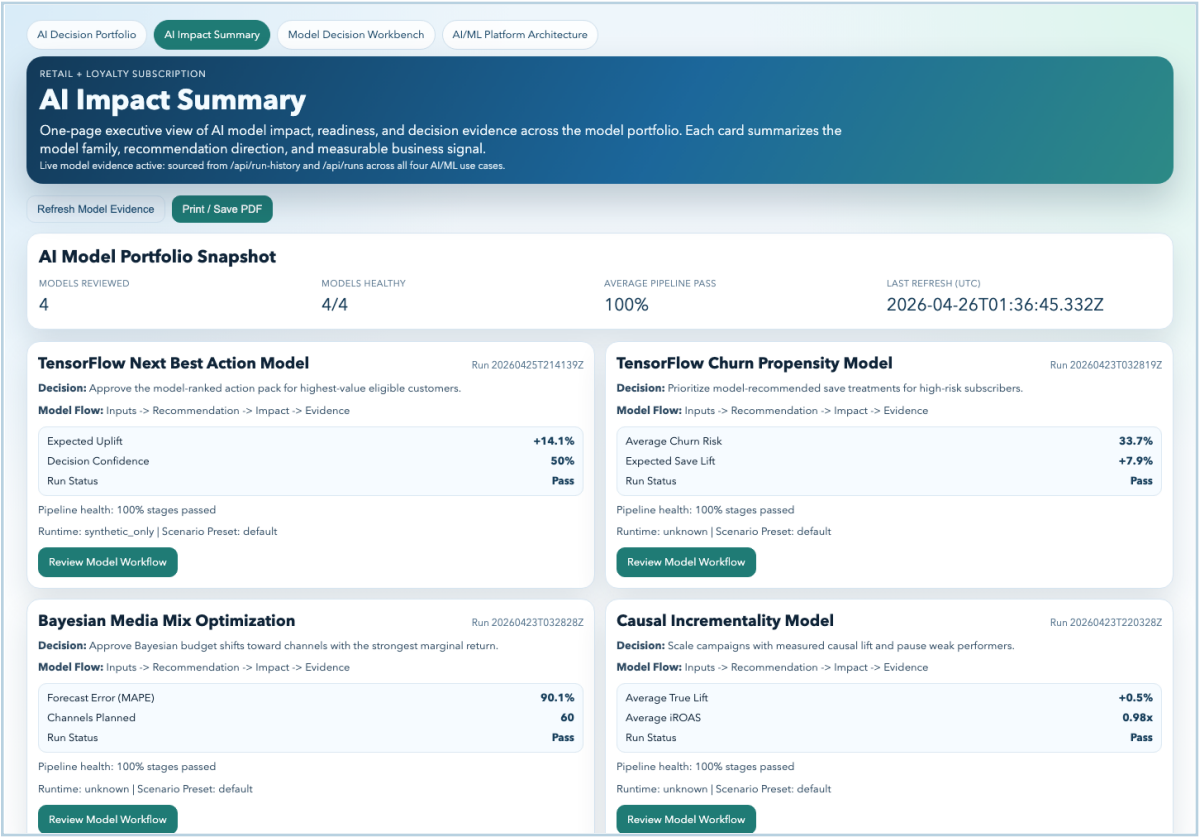
What to point out:

- Expected uplift summarizes predicted business value.
- Decision confidence shows the average model confidence signal.
- Activation records show that outputs are ready for downstream decisioning.

Flow Step 4: Review AI Impact Summary

Open AI Impact Summary when you need a portfolio-level executive view.

AI Impact Summary



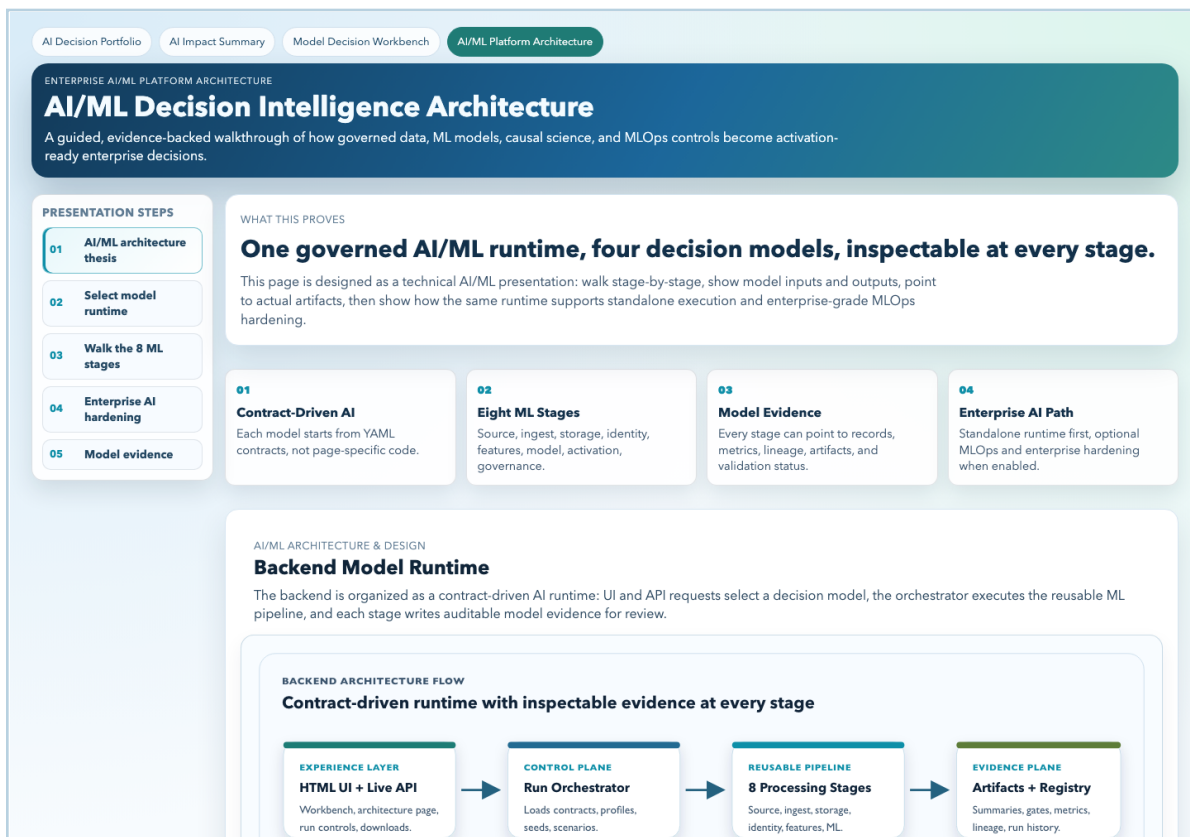
What to point out:

- Models reviewed and models healthy.
- Average pipeline pass rate.
- Model-level decisions, KPIs, and latest run status.

Flow Step 5: Explain The AI/ML Architecture

Open AI/ML Platform Architecture to show the technical thesis, backend model runtime, and component diagram.

AI/ML Architecture Thesis



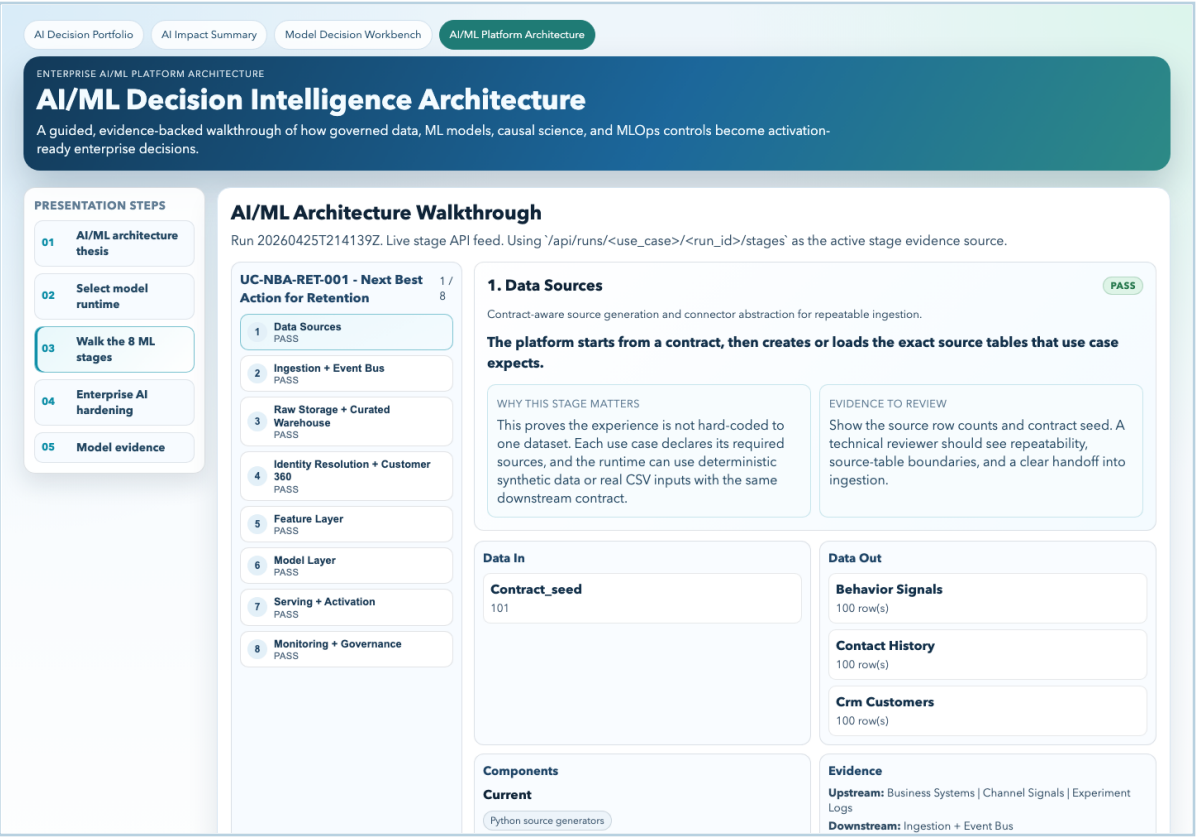
What to point out:

- One governed AI/ML runtime supports multiple decision models.
- The backend is contract-driven and evidence-backed.
- The architecture is designed for standalone runtime first, with optional enterprise integration.

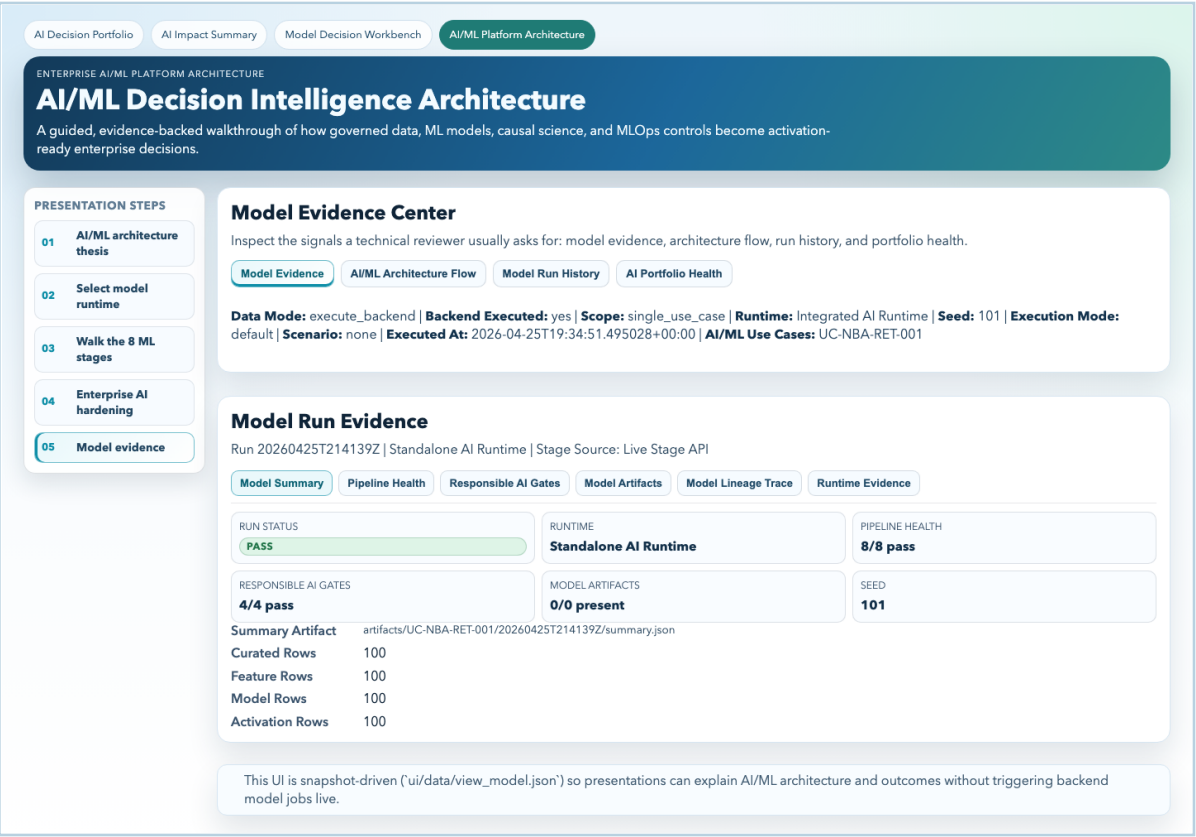
Flow Step 6: Walk The Eight ML Stages

Use the architecture walkthrough to explain the pipeline from Data Sources to Monitoring + Governance.

Eight ML Stages



Model Evidence Center



What to point out:

- Model Evidence is the first evidence tab.
- Architecture Flow explains the execution order.
- Model Run History compares recent runs.
- AI Portfolio Health summarizes latest run status across use cases.

Flow Step 8: Switch To Technical Model Evidence

In the Model Decision Workbench, switch from Decision View to Model Evidence for a deeper technical review.

Technical Model Evidence

The screenshot displays the 'Model Decision Workbench' interface for a 'RETAIL + LOYALTY SUBSCRIPTION' use case. The main title is 'Model Decision Workbench' with a subtitle: 'Inspect how AI/ML models convert governed inputs into recommendations, impact forecasts, decision evidence, and activation-ready outputs.'

Navigation: AI Decision Portfolio | AI Impact Summary | **Model Decision Workbench** | AI/ML Platform Architecture

Left Sidebar:

- AI MODEL:** TensorFlow NBA (selected), TensorFlow Churn, Bayesian MMM, Causal Lift
- EVIDENCE VIEW:** Decision View, **Model Evidence** (selected)
- STEP:** 1. Data + Model Contract, **2. Model Runtime + Activation** (selected), 3. Responsible AI Evidence

Main Content Area:

MODEL EVIDENCE VIEW
TensorFlow Next Best Action Model
Model evidence active from run, stage, governance, and artifact APIs.

Step 2: Model Runtime + Activation
Deep technical walkthrough: contracts, stage evidence, model lineage, and governance artifacts.

Model + Activation Detail
Inspect model backend, training split reliability, and activation-ready records.

MODEL BACKEND	MODEL VERSION	TRAIN/HOLDOUT
tensorflow Configured runtime backend	nba_retention_tf_v1 Versioned model lineage	80/20 Data split row counts

Model Metrics (Primitive)
Compact metric table from latest summary payload.

Metric	Value
primary_kpi	30_day_incremental_retention_lift
avg_expected_uplift	0.1411
avg_confidence	0.5011
model_backend	tensorflow
train_rows	80
holdout_rows	20
tf_train_loss	1.2867
tf_train_action_acc	0.3875
tf_holdout_loss	1.4645
tf_holdout_action_acc	0.1000

Right Sidebar:

- Model Run Snapshot**
Run ID: 20260425T214139Z
Status: **PASS**
Stage Pass: **100%**
Runtime: **synthetic_only**
- Model Evidence Snapshot**
Components: **8/8 pass**
Artifacts: **17/28 present**
Data Quality: **not_applicable (0/0)**
Model Backend: **tensorflow**
- Quick Actions**
[Open Model Workflow](#)
[Open AI Impact Summary](#)

What to point out:

- Contract controls show use case, runtime, scenario, and model constraints.
- Technical evidence connects components, APIs, stage outputs, and artifacts.
- The same workflow can be explained to business users or technical users without changing apps.

4. Starting The App

From the repository root:

```
make standalone
```

Open:

```
http://127.0.0.1:8080/ui/experience/index.html
```

If the environment is already prepared:

```
make standalone-fast
```

For live UI/API mode:

```
make ui-live
```

5. Main Pages

Page | URL | Primary Audience

AI Decision Portfolio | /ui/experience/business-home.html | Business and executive users

AI Impact Summary | /ui/experience/business-exec-summary.html | Executives and business owners

Model Decision Workbench | /ui/experience/workbench.html | Business users and technical reviewers

AI/ML Platform Architecture | /ui/experience/index.html | Architects, ML engineers, and platform teams

6. Business User Guide

Business users should focus on the decision story and model impact.

Step 1: Open AI Decision Portfolio

Use the portfolio page to understand which models are available and what each model is designed to decide.

Look for:

- Model name
- Decision output
- Primary business metric
- Recommended presentation sequence

Step 2: Open A Model Workflow

Click Open Model Workflow for one of the models.

Recommended first workflow:

TensorFlow Next Best Action Model

Step 3: Walk The Business Steps

In the Model Decision Workbench, use the left-side step navigation:

- Model Inputs
- Model Recommendation
- Impact Forecast
- Decision Evidence

Use these questions while presenting:

- What customer, campaign, or spend signals entered the model?
- What does the model recommend?
- What impact does the model forecast?
- What evidence supports the decision?

Step 4: Review The AI Impact Summary

Open AI Impact Summary to see portfolio-level model health.

Look for:

- Models reviewed
- Models healthy
- Average pipeline pass rate
- Latest run status

- Model-specific business KPIs

Step 5: Share Or Print

Use Print / Save PDF on the AI Impact Summary page when you need an executive-ready output.

7. Technical User Guide

Technical users should focus on architecture, run evidence, artifacts, and governance.

Step 1: Open AI/ML Platform Architecture

Use the architecture page to explain the end-to-end platform.

The core architecture is:

Data Sources

- > Ingestion + Event Bus
- > Raw Storage + Curated Warehouse
- > Identity Resolution + Customer 360
- > Feature Layer
- > Model Layer
- > Serving + Activation
- > Monitoring + Governance

Step 2: Walk The Eight ML Stages

Use the side navigation on the architecture page to walk stage by stage.

For each stage, review:

- What the stage does
- Data in
- Data out

- Components
- Evidence artifacts

Step 3: Use Model Evidence View

Open Model Decision Workbench and select Model Evidence.

Use this view to inspect:

- Contract controls
- Stage inputs and outputs
- Model backend
- Model metrics
- Activation payloads
- Validation gates
- Artifact paths

Step 4: Inspect Model Evidence Center

On the architecture page, open Model Evidence Center.

Review:

- Model Evidence
- AI/ML Architecture Flow
- Model Run History
- AI Portfolio Health

Step 5: Run A New Model Workflow

Use Model Execution controls on the architecture page.

Common options:

- Select an AI/ML use case.

- Choose Standalone AI Runtime or Integrated AI Runtime.
- Choose runtime mode.
- Select a scenario preset if needed.
- Click Run Standalone AI Runtime or Run Integrated AI Runtime.

Step 6: Simulate Stage Execution

Use the simulation controls to explain the pipeline gradually:

- Run Next Stage
- Pause Simulation
- Resume Simulation
- Reset Simulation

This is useful when presenting the architecture to technical audiences.

8. Runtime Profiles

Profile | Use When

Standalone AI Runtime | You want the app to run locally without external services.

Integrated AI Runtime | You want to show optional Kafka, Postgres, or MinIO-style infrastructure mirroring.

Enterprise Integration Profile | You want to show readiness for MLflow, Feast, Splink, Airflow, Keycloak, monitoring, and metrics.

The app can run standalone without deployment dependencies. Enterprise integrations are optional.

9. Common Tasks

Run all model workflows

```
python3 -m pipelines.cli run-stack-all
```

Run one workflow

```
python3 -m pipelines.cli run-stack --use-case UC-NBA-RET-001
```

Run synthetic-only mode

```
python3 -m pipelines.cli run-stack --use-case UC-NBA-RET-001 --runtime-mode  
synthetic_only
```

View recent runs

```
python3 -m pipelines.cli list-runs
```

Inspect run stages

```
python3 -m pipelines.cli show-stages --use-case UC-NBA-RET-001 --run-id <run_id>
```

Inspect model readiness

```
python3 -m pipelines.cli model-readiness --use-case UC-NBA-RET-001 --run-id <run_id>
```

10. Troubleshooting

If the app is not loading:

- Confirm the server is running.
- Confirm the URL uses the right port.
- Run make standalone-fast if dependencies and artifacts already exist.
- Run make smoke-standalone to validate the app.

If model evidence is missing:

- Run at least one workflow.

- Refresh the page.
- Check artifacts/<use_case_id>/<run_id>/summary.json.

If integrated runtime is unavailable:

- Use Standalone AI Runtime.
- Confirm optional infrastructure is running before using Integrated AI Runtime.

11. What To Show In A Meeting

For business stakeholders:

- AI Decision Portfolio
- One Model Workflow
- AI Impact Summary

For technical stakeholders:

- AI/ML Platform Architecture
- Eight ML stages
- Model Evidence View
- Model Run History
- Runtime Evidence

For platform leaders:

- Business value from model workflows
- Evidence-backed architecture
- Standalone runtime
- Enterprise integration path